

Gagan Bansal

bansalgagan2004@gmail.com — +91 9996892070 — LinkedIn — GitHub — LeetCode

Summary

A CS Graduate(CGPA 9.44, graduating June 2026) with R&D internship at Quark Software. Hands-on experience in backend systems, applied AI development, and cross-platform C++ porting.

Technical Skills

Languages: C++, Java, Python

Web and Backend: React.js, TailwindCSS, Node.js, Express.js, Apache Kafka, RESTful APIs

Databases: MongoDB, SQL, PostgreSQL

GenAI: LLM APIs, RAG Pipelines, Pinecone, Agentic AI, MCP, LangChain

Tools and Platforms: Figma, Git, Linux, macOS, Claude Code

Experience

Research and Development Intern, *Quark Software Inc.*

Jan 2026 – June 2026 *Mohali*

- Ported and modernized a legacy codebase to ensure compatibility with the latest macOS versions, supporting both ARM64 (Apple Silicon) and x86-64 architectures.
- Resolved complex cross-platform build and dependency issues involving OpenSSL and Axis2, ensuring stable builds across ARM64 and x86 architectures.
- Debugged and fixed low-level compatibility issues arising from architecture differences and deprecated system APIs.

Projects

oForge - Engineering Workspace Platform

demo

React.js, Node.js, Express.js, MongoDB, PostgreSQL, TailwindCSS

- Built oForge as a team collaboration platform for students, combining team discovery with a feature-rich workspace including Kanban boards, real-time chat, schema design, documentation, and whiteboard tools.
- Introduced a gated team collaboration system where users must submit proposals to join projects, with workspace access granted only after owner approval, ensuring privacy and controlled team formation.

RAG-based Multi-Document PDF Chatbot

GitHub

Python, LangChain, Pinecone, Google Gemini API, Streamlit, Retrieval-Augmented Generation (RAG)

- Built a multi-document PDF chatbot using RAG, enabling context-aware question answering with Google Gemini and semantic retrieval.
- Designed namespace-based indexing in Pinecone to isolate documents and support efficient, scalable multi-PDF querying.
- Implemented source attribution and optimized retrieval pipeline, improving response accuracy, transparency, and latency.

KVMemo – Multithreaded In-Memory Key-Value Store

demo

C++, TCP Sockets, Multithreading, Custom Thread Pool, Atomics

- Designed and implemented a high-performance in-memory key-value store in modern C++ supporting concurrent TCP client connections.
- Built a thread-pool architecture with task scheduling and synchronization using `std::mutex`, `std::shared-mutex`, and atomic counters to optimize read-heavy workloads.

Education

Chitkara University, Chandigarh

B.E Computer Science

Oct 2022 – June 2026

CGPA: 9.44

Achievements

- Solved **1300+** DSA problems across LeetCode, GFG, and CodeChef & **LeetCode Contest Rating: 1618**.
- Selected for UCA Batch at Chitkara University – Top 3% out of 2500+ students.
- Finalist** – Shopclues Smart AI Cataloging Hackathon (Top 5 of 70+ teams), Chandigarh, 2024.